

Light Field Cinematography (and the Birth of a Volumetrics Lab)

C K Olsen
Cinematic Arts
University of Iowa
Iowa City, Iowa, USA
ck@ckolsen.com



Figure 1: Volumetric Moving Image reconstruction featuring undergraduate researchers Isa Holtze, Calvin Czechin, and Fin Deevy, illustrating embodied performance rendered cinematically explorable through Light Field Cinematography.

Abstract

This talk examines an interdisciplinary faculty-led undergraduate research initiative at the University of Iowa exploring Light Field Cinematography (used here as a cinematic and computational framework for working with view-dependent, spatially navigable image fields) through the pedagogical development of a small-scale volumetrics lab. Emerging from ongoing visual narrative and performative explorations, the work integrates computer science, real-time graphics, cinematography, and dance performance to focus on extending Three-Dimensional Gaussian Splatting (3DGS) [Munkberg et al. 2023] into Volumetric Moving Images

(also known as Four-Dimensional Gaussian Splats or 4DGS). While such workflows are typically demonstrated in highly resourced technical environments, this project investigates how motion-based volumetric capture can be made legible, teachable, and creatively meaningful within undergraduate education. Our explorations have operated across a modular range of small- to mid-scale multi-camera configurations, using a variety of accessible capture devices and Gaussian Splatting workflows (e.g., Postshot, Supersplat, and Brush). This talk documents the role of iterative problem solving and experimentation in successfully implementing constraint-driven volumetric research within shared, resource-limited academic environments. These research methodologies position Light Field Cinematography as a pedagogical framework through which students collaboratively encounter the complexities of volumetric imaging across cinematic, photographic, performing arts, and computational disciplines, thus demonstrating how intention-driven creative, narrative, and performance-based volumetric capture processes enhance both engagement and technical refinement. This approach repositions



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

SIGGRAPH Educator's Forum '26, Los Angeles, CA, USA

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2552-4/26/07

<https://doi.org/10.1145/3799829.3812523>

core technical concerns (such as reconstruction fidelity) as mutable, creative parameters for exploring artistic inquiry and engaging with what I term Gaussian Narrativity, whereby narrative emerges through positional, temporal, and perspective-based engagement within moving, dimensional fields of captured light.

CCS Concepts

• **Computing methodologies** → **Computer graphics**; *Graphics systems and interfaces*; • **Human-centered computing**;

Keywords

Light Field Cinematography, volumetric imaging, Gaussian splatting, embodied performance, education

ACM Reference Format:

C K Olsen. 2026. Light Field Cinematography (and the Birth of a Volumetrics Lab). In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH Educator's Forum '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3799829.3812523>

1 Introduction

Recent advances in volumetric imaging have expanded the expressive possibilities of computer graphics, enabling representations that move beyond static three-dimensional form toward temporally rich, navigable fields of light and motion. Techniques such as Three-Dimensional Gaussian Splatting (3DGS) enable high-quality volumetric reconstructions, but the shift to Volumetric Moving Imagery is most often demonstrated within highly resourced research or commercial contexts. This talk presents an interdisciplinary educational initiative exploring Light Field Cinematography through the development of a small-scale volumetrics lab situated within a cinematic arts context. Emerging from ongoing visual narrative and performative explorations, the project investigates how volumetric moving imagery expands beyond technical reconstruction to function as a cinematic and perceptual mode of engagement with light, movement, gesture, and spatial presence. Undergraduate researchers collaborate across capture, reconstruction, and evaluation, engaging volumetric systems as both technical and artistic structures. Emphasizing clarity of process over algorithmic novelty, the work translates advanced volumetric techniques into moving image-based workflows that are legible, reproducible, and meaningful through creative and embodied inquiry. In doing so, Light Field Cinematography functions as a pedagogical framework through which students encounter volumetric imaging as an interdisciplinary cinematic practice, one that drives technical experimentation, refinement, and understanding within real-world educational constraints.

2 From 3DGS to Volumetric Moving Images

This project emphasizes the temporal extension of 3DGS into Volumetric Moving Images in which motion is translated into a continuously evolving field of dimensional light and form. For students, this reframes volumetric capture from a static modeling problem into a cinematically active one, where gesture, continuity, and temporal coherence are central. In Volumetric Moving Imagery, there are two cinematic modes: the originative capture camera configuration, and

the reconstructed (synthetic) light field camera. As such, it becomes critically important to consider both a scene's performative intention and its potential reconstruction fidelity. Approaches to this second mode of light field cinematography are ultimately shaped through the originative capture process itself, including intentional volumetric camera placement, framerate, shutter speed, lens choice, lighting design, and exposure, where each parameter directly influences the resulting volumetric field. Unlike contemporary 3DGS research which frequently prioritizes optimization and/or rendering performance, this work enables Volumetric Moving Imagery to serve as a pedagogical bridge between computational imaging and cinematic practice, supporting interdisciplinary learning without requiring deep prior technical specialization. Additional technical concerns, such as spatial and temporal registration, become critical forces that directly shape the coherence and fidelity of the reconstructed volumetric field, influencing the resolution of captured gesture, form, and overall perceptual continuity within the resulting light field camera space. Iterative experimentation with synchronization, camera-to-camera latency, and calibration methodologies therefore became essential to understanding how movement is both expressed and experienced volumetrically across time and space. While optimization remains an ongoing concern, these refinements ultimately function in support of creative and performative inquiry, where embodied movement and spatial engagement expose both the limitations and potential of the evolving system. These evolving relationships between movement, perception, and positional engagement begin to suggest what I term Gaussian Narrativity, in which narrative understanding emerges through spatial and temporal engagement with Volumetric Moving Imagery via the light field camera.



Figure 2: Student researchers preparing volumetric capture

3 Building a Modular Volumetrics Lab

The volumetrics lab has evolved around a core pedagogical constraint: workflows needed to remain reproducible, legible, and adaptable within undergraduate teaching environments. Originally conceived as a modular system that could be assembled, reconfigured, and iterated upon by students, the lab has to date employed reduced camera arrays, reconfigurable lighting, and accessible capture devices to support volumetric capture within shared studio spaces and limited production windows. Volumetric reconstruction

by necessity takes place using shared lab computing resources during off-hours, embedding iterative experimentation directly into the rhythms and logistical realities of the educational environment. During early research phases, this modular structure enabled repeated cycles of calibration, capture, teardown, data offload, reconstruction, and review within tightly constrained windows between scheduled classes. Camera placement, volume and lighting deployments, performer movement, spatial registration, and temporal alignment emerge from the work as dynamic and interconnected design variables, with each directly shaping the resulting volumetric field and influencing how movement and gesture are experienced across space and time. As the project evolves, embodied performance and creative experimentation increasingly inform technical refinement. Iterative investigations involving synchronization, camera-to-camera latency, calibration methodologies, and capture strategies continue to develop alongside performative inquiry, revealing how subtle shifts in temporal coherence, spatial continuity, and perspective alter perceptual engagement within the volumetric scene. Each capture session has thus become a compelling adaptive problem-solving environment operating under strict time-compressed constraints, forcing the team to contend with equipment failures, last-minute volume reconfigurations, unexpected parameter shifts, and calibration conditions requiring attention mid-session. As a result, these iterative cycles of capture, reconstruction, and refinement have enabled student researchers to collaboratively devise workflows that support successful volumetric capture under less-than-ideal circumstances. This transparency has positioned the lab as a shared site of cinematic, computational, and performative inquiry within an actively evolving pedagogical and creative research framework.

4 Pedagogical Outcomes and Implications

Grounding volumetric imaging within cinematic, embodied, and performative inquiry enables students from diverse disciplines to engage meaningfully with complex volumetric systems, despite their level of prior familiarity. Framing instruction through performance, lighting, spatial composition, temporality, and perceptual continuity supports conceptual understanding while encouraging students to encounter volumetric capture as an evolving cinematic process rather than a fixed technical workflow. Interdisciplinary collaboration has proven essential throughout the project, with students from engineering, computer science, cinema, and dance all contributing diverse insights to a collectively informed capture strategy, foregrounding performance design, spatial engagement, and evaluative interpretation. As iterative cycles of capture, reconstruction, and review unfold, students have encountered technical instability, reconstruction artifacts, synchronization issues, hardware limitations, and shifting calibration conditions as active components of the research process itself. These evolving conditions encourage adaptive problem-solving, collaborative evaluation, and critical discussion across disciplines, allowing students to participate directly in the refinement of both creative and technical methodologies. Embodied movement and performative experimentation have frequently exposed perceptual and cinematic challenges that might otherwise remain obscured within static reconstruction-focused

workflows, reinforcing the importance of gesture, continuity, temporal coherence, and spatial presence within Volumetric Moving Imagery. Taken together, this work demonstrates how the early development of a constraint-driven volumetrics lab can function as a powerful pedagogical and creative research framework. Through iterative experimentation and interdisciplinary collaboration, Light Field Cinematography enables students to engage volumetric imaging as a shared cinematic, computational, and performative practice grounded in inquiry, adaptation, and embodied exploration within real-world educational environments.

Acknowledgments

The author thanks the University of Iowa Office of Undergraduate Research (OUR), College of Liberal Arts and Sciences (CLAS), CLAS Production Services Unit, his undergraduate researchers (Calvin Czeschin, Tatum D' Emanuele, Fin Deevy, Gavin Egger, Isa Holtze, Vinnie Mazza, Nancy Nahra, Kristin To, Trinity Woodward, Nate Ingalls, and John Charlson), and the Departments of Computer Science, Cinematic Arts, and Dance.

References

- Jacob Munkberg, Jon Hasselgren, Alexander O. Ulusoy, Tianchang Yu, and Michael Vlastaras. 2023. Real-Time Radiance Fields Using Gaussian Splatting. *ACM Transactions on Graphics* 42, 4 (2023). doi:10.1145/3587424.3595585